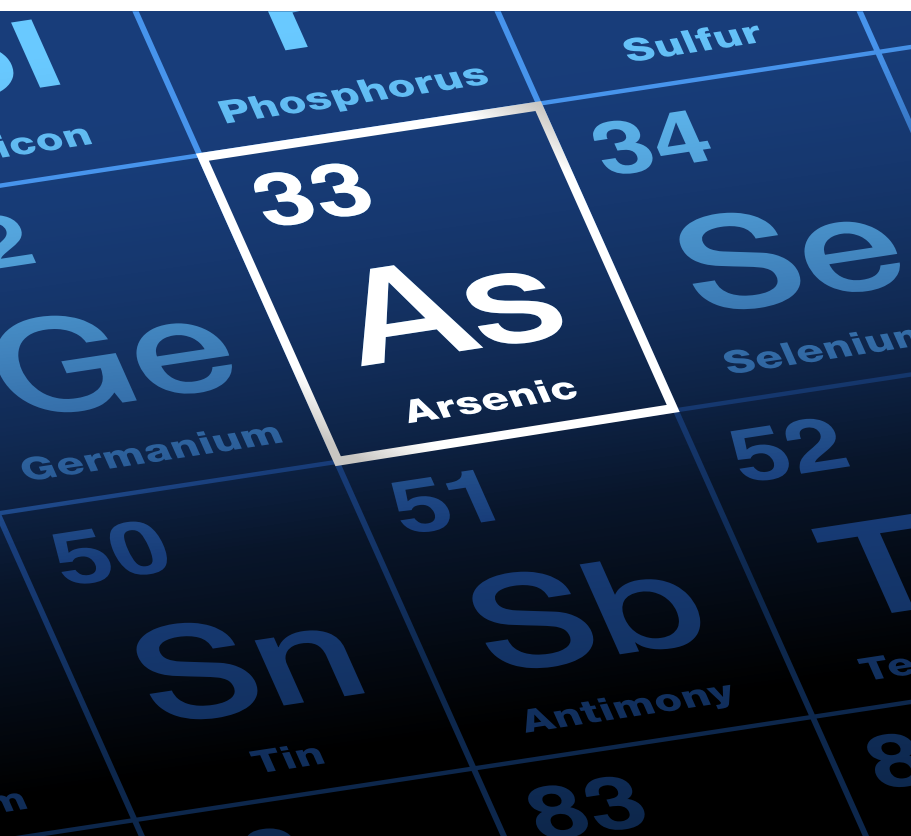


Tracking Regulated Inorganic Contaminant Level Violations

Katherine Alfredo and Natchaya Luangphairin



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Editor's note: This series of "Water Quality Matters" columns explores the innovation horizon, as experts from AWWA's 15 Water Quality Technology Division (WQTD) committees delve into emerging technologies and initiatives that are shaping the future of water quality. The series builds on previous WQTD columns that supported AWWA's Water 2050 initiative, which brought together diverse perspectives to "collaboratively imagine the water future we want to create." For more information, visit www.awwa.org/water2050.

While preparing this column, members of AWWA's Inorganic Contaminants Committee decided to reflect on how we, as a community of water professionals, have been doing over the past decade. To do this, we used the US Environmental Protection Agency's (EPA's) Safe

Drinking Water Information System violation database. The database is extremely important to communities in the United States, as it quarterly updates rule name, violation type, compliance, and compliance status for public water systems. US public water systems are those that provide potable water to at least 15 service connections or regularly serve at least 25 individuals for at least 60 days a year.

Compiling the data for the past decade (2014–2024) for inorganic contaminants listed under the Inorganic Rule, we restricted our analysis to active community water systems (CWSs) with maximum contaminant level (MCL) violations. A CWS is a public water system that supplies water to the same population year-round. We only considered unique public water system identifications (PWSIDs) for each year, avoiding any duplicate counting for each violation code. There are 14 regulated inorganic contaminants (antimony, arsenic, asbestos, barium, beryllium, cadmium, chromium, cyanide, fluoride, mercury, nitrate, nitrite, selenium, and thallium) with an MCL.

Positive Results

So how are we doing? In a word: steady, thanks to CWSs working hard to protect consumers from these contaminants. The data from the past decade show limited change in the number of utilities with a violation each year and the top contaminants of concern.

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Looking at data compiled in table 1, arsenic, nitrate, and fluoride are the top contaminants in violation of their respective MCLs. Nearly 20 years since the enforcement of EPA's revised arsenic rule in 2006, arsenic is still leading MCL violations, with more than 225 utilities in violation each year in the past decade (802 unique CWSs for the 10-year period). These utilities are primarily concentrated in California and Texas and are mostly composed of the smallest groundwater utilities (<3,301 population served), as shown in figure 1. Since 2006, more than 3,300 arsenic violations have been recorded across all categories (MCL, monitoring, and treatment technique). However, only 802 unique CWSs were responsible for MCL violations; of those, 733, or more than 90%, were small groundwater systems, highlighting the compliance challenges facing these smaller groundwater utilities.

As we reach the 20-year anniversary of the arsenic rule enforcement in the United States, tracking its progress has become a priority for AWWA's Inorganic Contaminants Committee.

Consider the 802 systems with arsenic MCL violations to the 49,387 total active CWSs in 2025, with 39,840 serving less than 3,300. These arsenic violations steadily increased from 49 CWSs in 2006, to a peak of 325 utilities being in violation in 2015, most likely as water professionals

Inorganic Contaminant MCL Violations by Contaminant and Year, 2014–2024

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Total Unique CWSs With Violations
Antimony		1	3	1	1		1		1	2	4	11
Arsenic	234	325	291	278	267	254	246	230	240	231	226	802
Asbestos	1		1	2	2	1	3	2	2	2	3	8
Barium	2	2	1	1	3	2	2	1		1	3	13
Beryllium		1		1		1	2	1	2	2	3	6
Cadmium		2	3	1	2	2				1		6
Chromium									1	1		1
Cyanide				1	1	1	1		1			2
Fluoride	37	53	46	48	41	40	41	36	41	50	45	111
Mercury	1				1	1						2
Nitrate	109	150	130	141	136	150	148	147	140	171	148	508
Nitrite	4	5	3	4	5	4	3	3	4	6	3	32
Selenium	7	7	9	5	4	7	10	9	5	6	6	27
Thallium			2					1	2		3	7
Total Unique CWSs with Violations	366	504	454	447	428	436	432	407	416	444	417	1433

CWS—community water system, MCL—maximum contaminant level

The totals are total number of unique CWSs in MCL violation, not the total number of violations.

Table 1

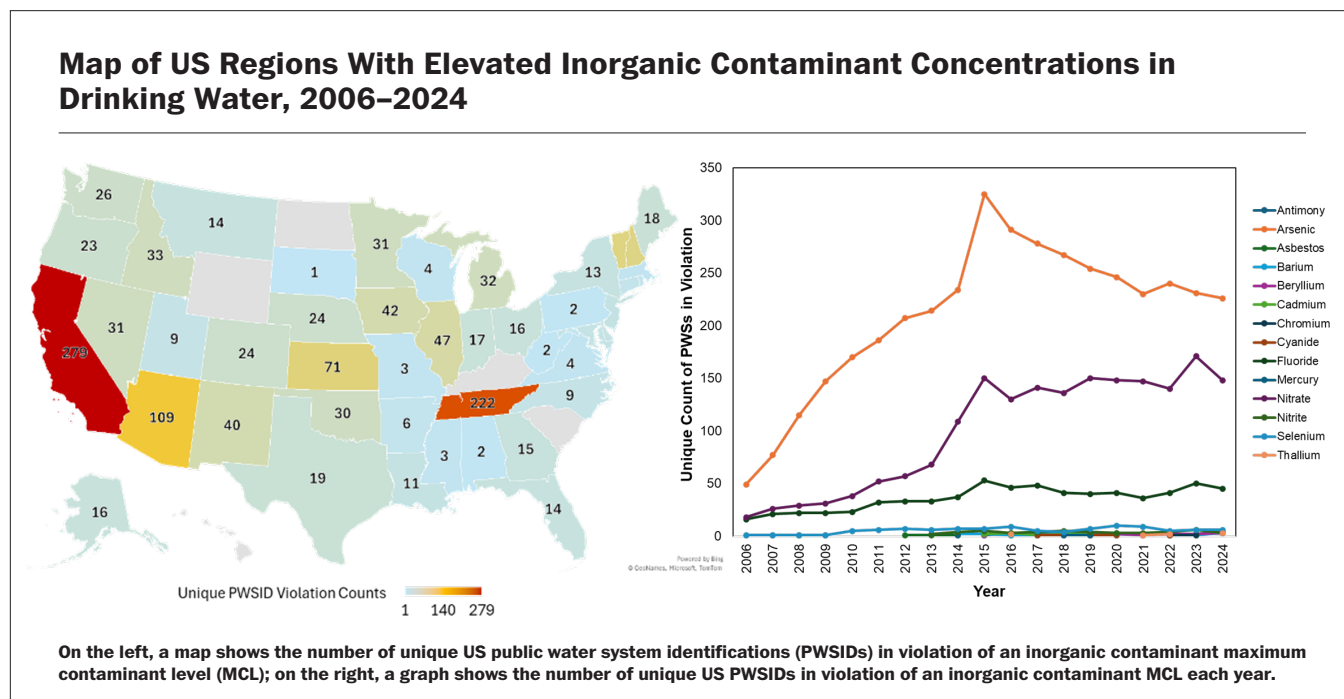


Figure 1

As more contaminants gain attention and new regulations are enacted, we must continue to support utilities that are dealing with arsenic contamination.

optimized treatment. The small fraction of utilities that are in violation in any given year (0.5% in 2024) are the ones that need additional support and training so they can continuously meet the primary standards.

The second most common violation is EPA's nitrate MCL, with a steady average of approximately 140 utilities a year reporting violations (508 CWSs for the 10-year period), followed by fluoride with only 111 CWS posting violations for that 10-year period. Aside from these contaminants, there are few utilities with MCL violations for the other major inorganic contaminants.

A Committee Agenda

As we reach the 20-year anniversary of the arsenic rule enforcement in the United States, tracking its progress has become a priority for AWWA's Inorganic Contaminants Committee. The committee launched an ambitious

initiative to take a retrospective look at the EPA's 50 arsenic demonstration sites to quantify cost and efficiency (by treatment and unique media). Despite detailed reports from these 50 sites, there is little information on what has happened since that original project ended, so we're trying to fill in the timeline from then to now. As more contaminants gain attention and new regulations are enacted, we must continue to support utilities that are dealing with arsenic contamination. If your utility has arsenic contamination challenges and/or you would like to help expand our inorganic contaminant database, please reach out to the committee's chair, Katherine Alfredo (kalfredo@usf.edu).

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<https://doi.org/10.1002/awwa.70060>